DERIVED TENSOR WITH A COMPACT FAITHFUL BIMODULE IS FULLY FAITHFUL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: assumption (2) is the substantive input and is used exactly once, at the shift case; dependencies carried as navigation stubs: Lemma lemma-tensor-hom-adjoint (adjunction), Lemma lemma-derived-restriction-exts (Ext interpretation), Remark remark-P-resolution and Remark remark-shift-tensor-no-sign, all in-source; the adjoint fully-faithful criterion is standard external tooling.

Editorial note. The label lemma-tensor-with-compact-fully-faithful names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Tensor-Hom adjunction. Same unit, Lemma lemma-tensor-hom-adjoint (notation and assumptions as stated there). The proof uses that the functor - tensor_A^L N has left adjoint RHom(N, -), given by Lemma lemma-tensor-hom-adjoint.
- Property (P) resolution induction. Same unit, Remark remark-P-resolution and Remark remark-shift-tensor-no-sign. The reduction to M = A[k] uses the property (P) resolution technique of Remark remark-P-resolution, and the identification A[k] tensor_A^L N = N[k] uses Remark remark-shift-tensor-no-sign.

Source reference. Original lemma 07RB: sources/stacks-chow/tags.txt, source lines 9861–9861. Prerequisite pointer invoked at proof line(s) 5700; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CS5: sources/stacks-dga/pinned-original-source.tex, source lines 5354–5354. Prerequisite pointer invoked at proof line(s) 5711; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09LT: sources/stacks-dga/pinned-original-source.tex, source lines 5633–5633. Prerequisite pointer invoked at proof line(s) 5687; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:dga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 09PB: sources/stacks-dga/pinned-original-source.tex, source lines 2953–2953. Prerequisite pointer invoked at proof line(s) 5713; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0FQN: sources/stacks-dga/pinned-original-source.tex, source lines 4998–4998. Prerequisite pointer invoked at proof line(s) 5716; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With notation and assumptions as in Lemma 09LT. Assume*

(1) *N defines a compact object of $D(B, d)$, and*
(2) *the map $H^k(A) \to \mathrm{Hom}_{D(B,d)}(N, N[k])$ is an isomorphism for all $k \in \mathbf{Z}$.*

Then the functor $- \otimes^{\mathbf{L}}_A N$ is fully faithful.

Proof. Our functor has a left adjoint given by $R\,\mathrm{Hom}(N, -)$ by Lemma 09LT. By Categories, Lemma 07RB it suffices to show that for a differential graded A-module M the map

$$M \longrightarrow R\,\mathrm{Hom}(N, M \otimes^{\mathbf{L}}_A N)$$

is an isomorphism in $D(A, \mathrm{d})$. For this it suffices to show that

$$H^n(M) \longrightarrow \mathrm{Ext}^n_{D(B,\mathrm{d})}(N, M \otimes^{\mathbf{L}}_A N)$$

is an isomorphism, see Lemma 0CS5. Since N is a compact object the right hand side commutes with direct sums. Thus by Remark 09PB it suffices to prove this map is an isomorphism for $M = A[k]$. Since $(A[k] \otimes^{\mathbf{L}}_A N) = N[k]$ by Remark 0FQN, assumption (2) on N is that the result holds for these. □